



Python for Data Science - 2305CS303

Lab - 3

Roll No. : 122

Name : Yatharth Sanghavi

1. WAP to print 1 to 10.

```
In [1]: for i in range(1,11):  
        print(i)
```

1
2
3
4
5
6
7
8
9
10

2. WAP to print 1 to n.

```
In [3]: n=int(input("enter n: "))  
        for i in range(1,n+1):  
            print(i)
```

1
2
3
4
5
6
7
8
9
10

3.WAP to print odd numbers between 1 to n.

```
In [2]: n=int(input("enter n: "))
        for i in range(1,n+1,2):
            print(i)
```

1
3
5
7
9

4. WAP to print numbers between two given numbers which is divisible by 2 but not divisible by 3.

```
In [4]: n1=int(input("enter n: "))
        n2=int(input("enter n2: "))
        for i in range(n1,n2+1):
            if i%2==0 and i%3!=0:
                print(i)
```

10
14
16
20

5. WAP to print sum of 1 to n numbers.

```
In [8]: n1=int(input("enter n: "))
        sum=0
        for i in range(1,n1+1):
            sum+=i
        print(sum)
```

55

6.WAP to print sum of series 1 + 4 + 9 + 16 + 25 + 36 + ...n.

```
In [10]: n1=int(input("enter n: "))
        sum=0
        for i in range(1,n1+1):
            sum+=i*i
        print(sum)
```

5

7. WAP to print sum of series 1 – 2 + 3 – 4 + 5 – 6 + 7 ... n.

```
In [11]: n1=int(input("enter n: "))
        sum=0
        for i in range(1,n1+1):
            if i%2==0:
                sum-=i
            else:
                sum+=i
        print(sum)
```

4

8. WAP to print multiplication table of given number.

```
In [12]: n1=int(input("enter n: "))
         for i in range(1,11):
             print(f"{n1} * {i}= {n1*i}")
```

```
10 * 1= 10
10 * 2= 20
10 * 3= 30
10 * 4= 40
10 * 5= 50
10 * 6= 60
10 * 7= 70
10 * 8= 80
10 * 9= 90
10 * 10= 100
```

9. WAP to find factorial of the given number.

```
In [13]: n1=int(input("enter n: "))
         sum=1
         for i in range(n1,1,-1):
             sum*=i
         print(sum)
```

```
120
```

10. WAP to find factors of the given number.

```
In [14]: n1=int(input("enter n: "))
         for i in range(1,n1+1):
             if n1%i==0:
                 print(i)
```

```
1
2
5
10
```

11. WAP to find whether the given number is prime or not.

```
In [16]: n1=int(input("enter n: "))
         # flag=True
         for i in range(2,n1):
             if n1%i==0:
                 print("not prime number")
                 break
         else:
             print("prime number")
```

```
prime number
```

12. WAP to print sum of digits of given number.

```
In [18]: n1=int(input("enter n: "))
         sum1=0
```

```
while n1>0:
    digit=n1%10
    sum1+=digit
    n1//=10
print(int(sum1))
```

6

13. WAP to check whether the given number is palindrome or not.

```
In [37]: n1=input("enter n: ")
n2=int(n1[::-1])
n1=int(n1)
if n1==n2:
    print("palindrome")
```

palindrome

14. Fibonacci Number

```
In [64]: n1=int(input("enter n: "))
a, b = 0, 1
for i in range(n1):
    print(a)
    a, b = b, a + b
```

0

1

1

2

3

15. Armstrong Number

```
In [73]: n1=int(input("enter n: "))
l=len(str(n1))
sum=0
t=n1
while n1>0:
    div=n1%10
    sum+= div**l
    n1//=10
if sum == t:
    print("it is Armstrong num")
else:
    print("not armstrong num")
```

it is Armstrong num

16. Perfect number

```
In [77]: n1=int(input("enter n: "))
sum=0
for i in range(1,n1):
    if n1%i==0:
        sum+=i
if sum==n1:
```

```
    print("perfect number")
else:
    print("not perfect number")
```

perfect number

17. GCD

```
In [80]: n1=int(input("enter n: "))
n2=int(input("enter n2: "))
for i in range(1,min(n1,n2)):
    if n1%i==0 and n2%i==0:
        max=i
print(max)
```

4

18. LCM

```
In [81]: n1=int(input("enter n: "))
n2=int(input("enter n2: "))
for i in range(1,min(n1,n2)):
    if n1%i==0 and n2%i==0:
        max=i
print((n1*n2)//max)
```

24

19. count no of digit

```
In [83]: n1=int(input("enter n: "))
print(len(str(n1)))
```

12

20.largest digit

```
In [86]: num = int(input("enter n: "))
digits = list(map(int, str(num)))
digits.sort()
print(digits[-1])
```

3